

Some other Open Source prototyping platforms

- Comet (an Indian Arduino clone)
- Pinguino (a PIC-based hardware)
- Arduino/Maple (an STM32-based hardware)
- Beagleboard
- PandaBoard
- Ustick51
- Induino
- chipKIT (a PIC 32-bit MCU-based Arduino solution from Microchip)
- Freeduino
- LilyPad
- Sanguino
- BeagleBone

simplicity of use that the platform offers. But this alone can't be the reason behind it," feels Jadhao. He adds, "The Arduino community has maintained and popularised the platform properly."

"Arduino forums are an excellent means of communication amongst the community that not only help people but are also a rich source of new ideas and thoughts to build better and more advanced hardware and software for the future," says Ram.

"In addition, many 'how to' guides and project ideas are available in the public domain. The design ideas and users' experiences are also shared in the community, which makes it easy even for beginners to work on the platform," informs Jadhao.

"Also, the Arduino community brings in a diverse range of used cases across various levels, and it is amazing to see the community being able to support Arduino consistently with a good level of quality. It is the community that makes Open Source projects successful over time," adds Ram.

5 Low cost and easy accessibility of the hardware

With ever-increasing affordability that the silicon vendors are adding these days, Arduino is becoming cheaper and more affordable to people. Starting at prices as low as Rs 500 (and

even lesser for some boards), Arduino enjoys immense cost advantage over other proprietary and Open Source boards.

"Not just in India, but all around the world Arduino is competitively priced as against proprietary boards. Another factor for its widespread adoption is the vast distribution network that Arduino has set up in India, making the boards' and shields' accessibility easy," says Chatterjee.

"In comparison, PandaBoard and BeagleBoard, which are also Open Source platforms (supported by Texas Instruments), are still not as easily available," shares Theo.

6 Open hardware and software

Arduino is completely Open Source both in terms of software as well as hardware. Both the hardware development board as well as the IDE are open for people to experiment as well as add/remove features as required.

When we say Arduino hardware is open in nature, it means that the schematics as well as building instructions can be used to build hardware without the need to buy from a particular vendor. Also, features can be added or removed in the existing design as per requirement. One can build the Arduino board by hand or use a local fab/hackerspace. Lots of such options are springing up slowly in India as Arduino is easy to clone too.

"This has proved to be a significant boon for the evolution of Arduino as a popular platform, as users are free to build varying versions of the hardware based on their needs and budget. For instance, in the case of hardware, users can remove the USB support if it is not required and manage with the serial port interface.

An on-board battery unit can also be added if a portable application has to be built. The regulator on board can also be changed to make it compatible with industrial-grade voltages," says Ram.

"The same is the case with software. You can use the standard Arduino IDE or configure to use the Eclipse IDE, Wiring IDE or graphical design software such as Scratch to program Arduino, which eliminates the need to write code. There are more projects of this kind in progress that would show up in future," Ram adds.

7 Arduino licencing policies are flexible and open

Creative commons attribution (CCA), general public licence (GPL) and lesser general public licence (LGPL) mainly support open hardware and allow designers to freely share designs. As regards Arduino, some excerpts from its licensing policy are shared below:

"Embedding an Arduino board inside a commercial product does not require you to disclose or open-source any information about its design. This would give the designer a complete control of the intellectual property that he has created."

"The source code for the Arduino environment is covered by the GPL, which requires any modifications to be open-sourced under the same license. It does not prevent the sale of derivative software or its inclusion in commercial products."

"Using the Arduino core and libraries for the firmware of a commercial product does not require you to release the source code for the firmware. The LGPL does, however, require you to make available object files that allow for the relinking of the firmware against updated versions of the Arduino core and libraries. Any modifications to the core and libraries must be released under the LGPL."

Libraries and reference designs will make it easy for designers to integrate Arduino into their electronics projects and vendors to clone these and sell variants